

CAP-ROUGE, QC, Canada

WMO: 711860

Lat: 48.374N

Lon: 70.535W

Elev: 24

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.0	-12.0	-23.8	1.5	-14.9	-19.7	1.9	-11.2	43.9	18.9	40.6	14.2	11.7	300	0.741

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	81.7	66.0	78.3	65.3	75.2	64.0	69.5	77.0	67.7	74.3	66.0	72.2	10.5	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.2	100.1	72.4	65.3	93.6	70.7	63.6	88.1	69.1	33.6	77.3	32.1	74.4	30.8	72.3	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
37.7	34.4	30.3	DB	-22.1	88.4	4.8	2.4	-25.5	90.0	-28.3	91.4	-30.9	92.8	-34.4	94.5
			WB	-22.8	73.0	4.7	2.5	-26.2	74.9	-28.9	76.3	-31.5	77.8	-34.9	79.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.0	8.8	11.2	23.5	37.4	51.2	61.4	66.2	65.1	56.9	45.0	32.8	18.8
	DBStd	21.71	12.59	11.66	11.35	7.32	6.93	6.36	4.79	5.11	6.45	6.54	7.66	11.07
	HDD50	5315	1276	1088	821	381	67	2	0	0	13	182	516	969
	HDD65	9316	1741	1508	1286	829	430	144	43	62	253	621	965	1434
	CDD50	1672	0	0	0	2	106	344	502	469	221	27	1	0
	CDD65	195	0	0	0	0	3	36	80	66	10	0	0	0
	CDH74	921	0	0	0	0	47	237	353	244	39	1	0	0
	CDH80	173	0	0	0	0	11	50	66	40	6	0	0	0
Wind	WSAvg	12.9	15.2	14.4	14.3	14.3	12.6	11.3	9.9	9.4	10.9	13.7	14.6	14.8
Precipitation	PrecAvg	37.7	2.2	2.1	2.0	2.2	3.2	3.9	4.5	4.1	3.9	3.7	2.8	3.0
	PrecMax	45.9	4.5	4.3	4.1	3.7	6.4	6.2	10.5	9.4	8.6	6.1	5.4	5.6
	PrecMin	31.9	0.3	0.9	0.7	0.8	1.3	1.7	1.5	1.6	2.1	1.1	0.9	1.5
	PrecStd	3.6	0.9	0.9	0.9	0.8	1.2	1.2	1.9	1.7	1.3	1.3	0.9	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.0	38.4	47.7	62.0	80.5	85.7	86.0	84.6	78.7	66.9	55.5	40.8
		MCWB	36.5	35.6	39.3	47.9	61.2	66.8	69.4	69.0	66.6	58.3	50.8	37.8
	2%	DB	34.7	34.7	42.8	55.0	71.8	80.1	81.3	80.0	73.0	60.4	48.0	37.2
		MCWB	33.6	32.7	36.7	44.0	55.9	63.8	66.8	67.4	63.8	53.8	44.7	35.3
	5%	DB	31.0	32.2	39.6	50.4	67.0	76.1	77.9	76.3	69.2	56.8	44.9	34.8
		MCWB	29.8	30.7	35.2	41.6	53.6	62.0	65.8	65.6	61.5	51.7	42.2	33.1
	10%	DB	26.5	29.2	37.1	46.6	62.3	72.1	74.7	73.4	65.9	53.5	42.2	32.9
		MCWB	25.2	27.9	33.8	39.8	51.7	60.3	64.6	64.3	59.7	49.6	39.4	31.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.0	36.4	41.4	48.9	63.3	69.9	72.2	71.9	69.0	59.4	52.0	38.3
		MCDB	38.9	37.8	44.7	56.7	76.4	80.9	78.5	78.8	74.8	65.1	54.6	40.3
	2%	WB	33.6	33.1	37.8	45.0	58.5	66.7	69.6	69.4	65.4	55.2	45.6	35.6
		MCDB	34.7	34.1	41.0	51.9	66.9	75.3	76.8	76.4	71.1	58.9	47.4	36.9
	5%	WB	30.1	30.7	35.8	42.6	55.5	64.6	67.8	67.5	62.8	52.5	42.5	33.4
		MCDB	31.2	32.1	38.8	48.1	64.1	72.1	74.3	73.7	67.7	55.8	44.7	34.5
	10%	WB	25.4	27.8	33.8	40.7	53.0	62.6	66.0	65.8	60.4	50.0	39.7	31.6
		MCDB	26.5	29.2	36.6	45.3	61.1	70.0	72.1	71.3	64.6	52.9	41.9	32.8
Mean Daily Temperature Range	5% DB	MDBR	13.9	16.1	15.0	13.7	17.1	16.6	14.7	14.2	13.1	9.9	8.2	10.0
		MCDBR	15.3	13.9	15.4	20.9	25.8	23.4	20.2	19.0	17.9	15.5	11.8	9.7
		MCWBR	14.6	12.9	11.5	12.4	12.4	10.4	9.0	9.1	10.2	10.6	9.8	9.2
	5% WB	MCDBR	15.7	13.7	13.7	17.6	21.9	18.9	16.9	15.9	16.5	14.3	11.3	9.5
		MCWBR	15.3	12.8	10.8	11.6	12.1	10.0	8.8	8.7	10.5	10.7	10.2	9.3
Clear-Sky Solar Irradiance	taub	0.268	0.290	0.323	0.395	0.374	0.395	0.418	0.397	0.381	0.338	0.313	0.274	
	taud	2.271	2.217	2.184	2.134	2.372	2.349	2.301	2.362	2.389	2.490	2.411	2.310	
	Ebn at Noon	254	273	281	270	281	275	266	267	260	256	236	236	
	Edh at Noon	23	31	38	45	37	38	39	36	31	24	21	20	
All-Sky Solar Radiation	RadAvg	415	721	1117	1413	1635	1745	1648	1401	1074	615	335	274	
	RadStd	51	95	92	110	157	117	161	93	81	62	34	29	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page